D1 Mensuration — Answer Key

IGCSE Mathematics 0580 | Full Solutions

D1_Q1: Perimeter of 2D Shapes — Problems

F Foundation — Answer: 26 cm

P = 2(l + w) = 2(8 + 5) = 2 x 13 = 26 cm

F Foundation — Answer: 24 cm

P = 4s = 4 x 6 = 24 cm

F Foundation — Answer: 21 cm

P = a + b + c = 7 + 5 + 9 = 21 cm

F Foundation — Answer: 44 cm

C = 2 x pi x r = 2 x (22/7) x 7 = 44 cm

C Core — Answer: 5 cm

P = 2(l + w), so 34 = 2(12 + w), 17 = 12 + w, w = 5 cm

C Core — Answer: 8 cm

A regular hexagon has 6 equal sides. s = 48 / 6 = 8 cm

C Core — Answer: 44 cm

C = pi x d = (22/7) x 14 = 44 cm

C Core — Answer: 36 cm

Arc = pi x r = (22/7) x 7 = 22 cm. Diameter = 14 cm. Total = 22 + 14 = 36 cm

C Core — Answer: 32 m

P = 10 + 6 + 4 + 3 + 6 + 3 = 32 m

C Core — Answer: 220 m

C = 2 x (22/7) x 35 = 220 cm per rotation. Distance = 220 x 100 = 22,000 cm = 220 m

C Core — Answer: 11 cm

Third side = 30 - 11 - 8 = 11 cm

C Core — Answer: 36 cm

P = 8 x 4.5 = 36 cm

H Challenge — Answer: (8x + 6) cm

P = 2(l + w) = 2((3x + 2) + (x + 1)) = 2(4x + 3) = 8x + 6 cm

H Challenge — Answer: 44.57 cm

Rectangle contributes 12 + 8 + 8 = 28 cm (one long side replaced by semicircle). Arc = pi x 4 = 12.568 cm. Plus other long side 12 cm. Wait: the semicircle replaces one 12 cm side. So P = 8 + 8 + 12 + (pi x 4) = 28 + 12.568 = 40.568 cm. Hmm, let me recalculate. The shape: rectangle 12 x 8. Semicircle on one 12 cm side (radius 6 cm). P = 12 + 8 + 8 + (pi x 6) = 28 + 18.852 = 46.852 cm. Let me use r=4 cm semicircle on one 8 cm side. P = 12 + 12 + 8 + (pi x 4) = 32 + 12.568 = 44.568 cm approx.

H Challenge — Answer: 90 degrees

Arc = (theta/360) x 2 x pi x r. 15.71 = (theta/360) x 62.84. theta/360 = 0.25. theta = 90 degrees.

H Challenge — Answer: 5x + 9 cm

Square contributes 3 sides: 3(x+3). Triangle contributes 2 sides: 2x. Shared side not counted. Total = 3x+9+2x = 5x+9.

D1_Q2: Area of 2D Shapes — Problems

F Foundation — Answer: 63 cm^2

A = l x w = 9 x 7 = 63 cm^2

F Foundation — Answer: 64 cm^2

A = s^2 = 8 x 8 = 64 cm^2

F Foundation — Answer: 30 cm^2

A = (1/2) x b x h = (1/2) x 12 x 5 = 30 cm^2

F Foundation — Answer: 154 cm^2

A = pi x r^2 = (22/7) x 7 x 7 = 154 cm^2

C Core — Answer: 60 cm^2

A = b x h = 10 x 6 = 60 cm^2

C Core — Answer: 50 cm^2

A = (1/2)(a + b)h = (1/2)(8 + 12) x 5 = (1/2) x 20 x 5 = 50 cm^2

C Core — Answer: 154 cm^2

r = 14/2 = 7 cm. A = pi x r^2 = (22/7) x 49 = 154 cm^2

C Core — Answer: 157.1 cm^2

A = (1/2) x pi x r^2 = 0.5 x 3.142 x 100 = 157.1 cm^2

C Core — Answer: 12 cm

A = (1/2) x b x h, so 42 = (1/2) x b x 7, 42 = 3.5b, b = 12 cm

C Core — Answer: 50,000 cm^2

1 m^2 = 10,000 cm^2. 5 m^2 = 5 x 10,000 = 50,000 cm^2

C Core — Answer: 12 cm

A = l x w, so 96 = l x 8, l = 96/8 = 12 cm

C Core — Answer: 500 cm^2

A = (1/2) x 40 x 25 = 500 cm^2

H Challenge — Answer: 12 cm

A = (1/2)(a + b)h. 60 = (1/2)(8 + b)(6). 60 = 3(8 + b). 20 = 8 + b. b = 12 cm

H Challenge — Answer: 4% decrease

Original area = 150 cm^2. New length = 18 cm. New width = 8 cm. New area = 144 cm^2. Change = (144-150)/150 x 100 = -4%.

H Challenge — Answer: 33.51 cm^2

A = (60/360) x pi x 64 = (1/6) x 3.142 x 64 = 33.515 cm^2

H Challenge — Answer: 6x + 8 cm^2

A = (1/2)((x+3)+(2x+1)) x 4 = 2(3x+4) = 6x+8 cm^2

D1_Q3: Composite Shapes — Problems

F Foundation — Answer: 26 m^2

A = 5 x 4 + 3 x 2 = 20 + 6 = 26 m^2

F Foundation — Answer: 33 m^2

A_rect = 6 x 4 = 24 m^2. A_tri = (1/2) x 6 x 3 = 9 m^2. Total = 33 m^2

F Foundation — Answer: 22 m

P = 7 + 4 + 3 + 2 + 4 + 2 = 22 m

F Foundation — Answer: 74 m^2

A_lawn = 10 x 8 = 80 m^2. A_bed = 3 x 2 = 6 m^2. Remaining = 80 - 6 = 74 m^2

C Core — Answer: 122.5 cm^2

A = pi(R^2 - r^2) = 3.142(64 - 25) = 3.142 x 39 = 122.538 cm^2

C Core — Answer: 74.14 m^2

A_rect = 10 x 6 = 60 m^2. A_semi = (1/2) x 3.142 x 9 = 14.139 m^2. Total = 74.139 m^2

C Core — Answer: 7 m

Total width = 12 m. Known horizontal = 5 m. Missing = 12 minus 5 = 7 m.

C Core — Answer: 46 m^2

Outer = 15 x 10 = 150 m^2. Inner = (15-2) x (10-2) = 13 x 8 = 104 m^2. Path = 150 - 104 = 46 m^2

C Core — Answer: 72 cm^2

A_rect = 48 cm^2. A_tri = (1/2) x 8 x 6 = 24 cm^2. Total = 72 cm^2

C Core — Answer: 36.85 cm

Two long sides = 12 + 12 = 24 cm. One short side = 5 cm replaced by semicircle arc. Arc = pi x 2.5 = 7.855 cm. Other short side = 5 cm. Total = 24 + 5 + 7.855 = 36.855 cm

C Core — Answer: 46 m^2

A_rect = 10 x 4 = 40 m^2. A_tri = (1/2) x 3 x 4 = 6 m^2. Total = 46 m^2

C Core — Answer: 5 m

Known sides sum = 15 + 8 + 7 + 3 + 8 = 41 m. Unknown = 46 - 41 = 5 m

H Challenge — Answer: 4x + (pi x^2)/8

A_rect = 4x. r = x/2, A_semi = (1/2) x pi x (x/2)^2 = (1/2) x pi x x^2/4 = (pi x^2)/8. Total = 4x + (pi x^2)/8

H Challenge — Answer: 21.45 cm^2

A_square = 100 cm^2. Four quarter-circles = one full circle of radius 5: A_circle = 3.142 x 25 = 78.55 cm^2. Remaining = 100 - 78.55 = 21.45 cm^2

H Challenge — Answer: 34.56 m^2

Outer radius = 6 m. A_path = pi(36-25) = 3.142 x 11 = 34.562 m^2

H Challenge — Answer: ab + cd

Total area = area of first rectangle + area of second rectangle = ab + cd

D1_Q4: Surface Area of Prisms — Problems

F Foundation — Answer: 94 cm^2

SA = 2(lw + lh + wh) = 2(5x4 + 5x3 + 4x3) = 2(20 + 15 + 12) = 94 cm^2

F Foundation — Answer: 216 cm^2

SA = 6s^2 = 6 x 36 = 216 cm^2

F Foundation — Answer: 408.46 cm^2

Circular ends: 2 x pi x 25 = 157.1 cm^2. Curved: 2 x pi x 5 x 8 = 251.36 cm^2. Total = 408.46 cm^2

F Foundation — Answer: 228 cm^2

Two triangles: 2 x (1/2 x 6 x 4) = 24 cm^2. Rectangles: 6x10 + 5x10 + 5x10 = 60 + 50 + 50 = 160 cm^2. Total = 24 + 160 = 184 cm^2. Wait: the slant sides - if triangle has base 6, height 4, the slant sides are sqrt(3^2+4^2)=5 cm each. So rectangles: 6x10 + 5x10 + 5x10 = 60+50+50=160. Two triangles: 2 x 12 = 24. Total = 184 cm^2.

C Core — Answer: 6 cm

214 = 2(7x5 + 7h + 5h) = 2(35 + 12h) = 70 + 24h. 144 = 24h. h = 6 cm

C Core — Answer: 301.63 cm^2

One circular end: pi x 16 = 50.272 cm^2. Curved surface: 2 x pi x 4 x 10 = 251.36 cm^2. Total = 301.632 cm^2

C Core — Answer: 1440 cm^2

SA = 2(20x15 + 20x12 + 15x12) = 2(300 + 240 + 180) = 1440 cm^2

C Core — Answer: 341 cm^2

Ends: 2 x (22/7) x 12.25 = 2 x 38.5 = 77 cm^2. Curved: 2 x (22/7) x 3.5 x 12 = 264 cm^2. Total = 341 cm^2

C Core — Answer: 120 cm^2

Hypotenuse = 5 cm. Two triangles: 2 x (1/2 x 3 x 4) = 12 cm^2. Rectangles: 3x8 + 4x8 + 5x8 = 24 + 32 + 40 = 96 cm^2. Total = 108 cm^2.

C Core — Answer: 4

Original SA = 6 x 9 = 54. New SA = 6 x 36 = 216. Factor = 216/54 = 4

C Core — Answer: 628.4 cm^2

r = 5 cm. Ends: 2 x pi x 25 = 157.1 cm^2. Curved: 2 x pi x 5 x 15 = 471.3 cm^2. Total = 628.4 cm^2

C Core — Answer: 188 cm^2

SA = 2 x cross-section area + perimeter x length = 2 x 22 + 24 x 6 = 44 + 144 = 188 cm^2

H Challenge — Answer: 2 pi r^2 + 1000/r

V = pi r^2 h = 500, so h = 500/(pi r^2). SA = 2 pi r^2 + 2 pi r h = 2 pi r^2 + 2 pi r x 500/(pi r^2) = 2 pi r^2 + 1000/r

H Challenge — Answer: 6x^2 + 2x - 4

SA = 2[(x+2)(x) + (x+2)(x-1) + x(x-1)] = 2[x^2+2x + x^2+2x-x-2 + x^2-x] = 2[3x^2 + 2x - 2] = 6x^2 + 4x - 4

H Challenge — Answer: 301.63 cm^2

Cylinder curved: 2 x pi x 4 x 6 = 150.82. Cylinder base circle: pi x 16 = 50.27. Hemisphere curved: 2 x pi x 16 = 100.54. Total = 150.82 + 50.27 + 100.54 = 301.63 cm^2

H Challenge — Answer: 6x^2 + 12x + 4

SA = 2[x(x+1) + x(x+2) + (x+1)(x+2)] = 2[x^2+x + x^2+2x + x^2+3x+2] = 2[3x^2+6x+2] = 6x^2+12x+4

D1_Q5: Volume of Prisms and Cylinders — Problems

F Foundation — Answer: 120 cm^3

V = l x w x h = 6 x 5 x 4 = 120 cm^3

F Foundation — Answer: 343 cm^3

V = s^3 = 7 x 7 x 7 = 343 cm^3

F Foundation — Answer: 502.72 cm^3

V = pi x r^2 x h = 3.142 x 16 x 10 = 502.72 cm^3

F Foundation — Answer: 240 cm^3

Area of triangle = (1/2) x 8 x 5 = 20 cm^2. V = 20 x 12 = 240 cm^3

C Core — Answer: 6 cm

V = l x w x h. 360 = 12 x 5 x h. 360 = 60h. h = 6 cm

C Core — Answer: 60 litres

V = 50 x 30 x 40 = 60,000 cm^3 = 60 litres

C Core — Answer: 3080 cm^3

r = 7 cm. V = (22/7) x 49 x 20 = 22 x 7 x 20 = 3080 cm^3

C Core — Answer: 225 cm^3

V = area of cross-section x length = 25 x 9 = 225 cm^3

C Core — Answer: 282.78 litres

V = 3.142 x 900 x 100 = 282,780 cm^3 = 282.78 litres

C Core — Answer: 375 m^3

V = 25 x 10 x 1.5 = 375 m^3

C Core — Answer: 5.15 cm

V = pi r^2 h. 1000 = 3.142 x r^2 x 12. r^2 = 1000/(3.142 x 12) = 26.53. r = sqrt(26.53) = 5.15 cm

C Core — Answer: 2,500,000 cm^3

1 m^3 = 1,000,000 cm^3. 2.5 m^3 = 2.5 x 1,000,000 = 2,500,000 cm^3

H Challenge — Answer: 904.78 cm^3

V_cyl = 3.142 x 36 x 12 = 1357.34 cm^3. V_hemi = (2/3) x 3.142 x 216 = 452.57 cm^3. Remaining = 1357.34 - 452.57 = 904.78 cm^3

H Challenge — Answer: 3x^2 + 14x + 8

V = A x L = (3x+2)(x+4) = 3x^2 + 12x + 2x + 8 = 3x^2 + 14x + 8 cm^3

H Challenge — Answer: 280 cm^3

Area of trapezium = (1/2)(5+9) x 4 = 28 cm^2. V = 28 x 10 = 280 cm^3

H Challenge — Answer: 113.1 litres

V = pi x 900 x 40 = 113,112 cm^3 = 113.1 litres

D1_Q6: Volume of Pyramids and Cones — Problems

F Foundation — Answer: 90 cm^3

V = (1/3) x l x w x h = (1/3) x 6 x 5 x 9 = (1/3) x 270 = 90 cm^3

F Foundation — Answer: 66 cm^3

V = (1/3) x pi x r^2 x h = (1/3) x (22/7) x 9 x 7 = (1/3) x 22 x 9 = 66 cm^3

F Foundation — Answer: 32 cm^3

Base area = 16 cm^2. V = (1/3) x 16 x 6 = 32 cm^3

F Foundation — Answer: 314.2 cm^3

r = 5 cm. V = (1/3) x 3.142 x 25 x 12 = (1/3) x 942.6 = 314.2 cm^3

C Core — Answer: 3 times

V_prism = 8x5x12 = 480 cm^3. V_pyramid = (1/3) x 480 = 160 cm^3. Ratio = 480/160 = 3

C Core — Answer: 3.82 cm

V = (1/3) pi r^2 h. 100 = (1/3) x 3.142 x 25 x h. 100 = 26.183h. h = 3.82 cm

C Core — Answer: 80 cm^3

Base area = (1/2) x 6 x 8 = 24 cm^2. V = (1/3) x 24 x 10 = 80 cm^3

C Core — Answer: 141.39 cm^3

V = (1/3) x 3.142 x 9 x 15 = (1/3) x 424.17 = 141.39 cm^3

C Core — Answer: 7.07 cm

V = (1/3) x s^2 x h. 200 = (1/3) x s^2 x 12 = 4s^2. s^2 = 50. s = sqrt(50) = 7.07 cm

C Core — Answer: 1465.1 cm^3

V_large = (1/3) x 3.142 x 81 x 18 = 1526.8 cm^3. V_small = (1/3) x 3.142 x 9 x 6 = 56.556 cm^3. V_frustum = 1526.8 - 56.556 = 1470.0 cm^3. Wait: V_large = (1/3) x pi x 81 x 18 = 486 x pi = 486 x 3.142 = 1527.0. V_small = (1/3) x pi x 9 x 6 = 18 x pi = 56.556. V_frustum = 1527.0 - 56.556 = 1470.444 cm^3.

C Core — Answer: 8

Original V = (1/3) pi x 16 x 9 = 48 pi. New: r=8, h=18. V = (1/3) pi x 64 x 18 = 384 pi. Ratio = 384/48 = 8

C Core — Answer: 9 cm

Base area = 60 cm^2. V = (1/3) x 60 x h = 20h. 180 = 20h. h = 9 cm

H Challenge — Answer: 12 cm

V_hemi = (2/3) pi r^3 = (2/3) pi x 216 = 144 pi. V_cone = (1/3) pi r^2 h = (1/3) pi x 36 x h = 12 pi h. 12 pi h = 144 pi. h = 12 cm

H Challenge — Answer: 489.06 cm^3

Full cone: h = 9 x 5/(5-2) = 15 cm. V_full = (1/3) pi x 25 x 15 = 125 pi = 392.75. Small cone: V_small = (1/3) pi x 4 x 6 = 8 pi = 25.136. Frustum = 392.75 - 25.136 = 367.614. Wait let me recalculate properly with similar triangles. Full cone: bottom r=5, top r=2, frustum h=9. Using similar triangles: H_full/5 = (H_full-9)/2. 2H = 5H - 45, 3H = 45, H=15. Top cone height = 6. V_full = (1/3) pi x 25 x 15 = 125 pi = 392.75. V_top = (1/3) pi x 4 x 6 = 8 pi = 25.136. Frustum = 392.75 - 25.136 = 367.614 cm^3

H Challenge — Answer: 1:8

V_A = (1/3) x 36 x 10 = 120. V_B = (1/3) x 144 x 20 = 960. Ratio = 120:960 = 1:8

H Challenge — Answer: 8/27

Full cone: r=5, h=15. When depth=10, the remaining water forms a smaller cone similar to the original. Height ratio = 10/15 = 2/3. Radius ratio = 2/3. Volume ratio = (2/3)^3 = 8/27.

D1_Q7: Volume of Spheres and Composite Solids — Problems

F Foundation — Answer: 904.90 cm^3

V = (4/3) x pi x r^3 = (4/3) x 3.142 x 216 = 904.896 cm^3

F Foundation — Answer: 56.56 cm^3

V = (2/3) x pi x r^3 = (2/3) x 3.142 x 27 = 56.556 cm^3

F Foundation — Answer: 314.2 cm^2

SA = 4 x pi x r^2 = 4 x 3.142 x 25 = 314.2 cm^2

F Foundation — Answer: 523.67 cm^3

r = 5 cm. V = (4/3) x 3.142 x 125 = 523.667 cm^3

C Core — Answer: 636.91 cm^3

V_cyl = 3.142 x 16 x 10 = 502.72 cm^3. V_hemi = (2/3) x 3.142 x 64 = 134.19 cm^3. Total = 636.91 cm^3

C Core — Answer: 6 cm

(4/3) pi r^3 = 288 pi. r^3 = 288 x 3/4 = 216. r = 6 cm

C Core — Answer: 273.35 cm^3

V_cyl = 3.142 x 9 x 8 = 226.22 cm^3. V_cone = (1/3) x 3.142 x 9 x 5 = 47.13 cm^3. Total = 273.35 cm^3

C Core — Answer: 27

Scale factor k = 9/3 = 3. Volume ratio = k^3 = 27

C Core — Answer: 134.04 cm^3

V_cyl = 3.142 x 4 x 8 = 100.544 cm^3. Two hemispheres = one sphere: V_sphere = (4/3) x 3.142 x 8 = 33.498 cm^3. Total = 134.042 cm^3

C Core — Answer: 32 cm

V_sphere = (4/3) pi x 216 = 288 pi. V_cyl = pi x 9 x h = 9 pi h. 9 pi h = 288 pi. h = 32 cm

C Core — Answer: 462 cm^2

Curved surface = 2 pi r^2 = 2 x (22/7) x 49 = 308 cm^2. Base = pi r^2 = (22/7) x 49 = 154 cm^2. Total = 462 cm^2

C Core — Answer: 14,139 litres

V = (4/3) x 3.142 x 3.375 = 14.139 m^3 = 14,139 litres

H Challenge — Answer: cube_root((3/4) r^2 h)

V_cyl = pi r^2 h. V_sphere = (4/3) pi R^3. Equate: (4/3) pi R^3 = pi r^2 h. R^3 = (3/4) r^2 h. R = cube_root(3r^2 h / 4)

H Challenge — Answer: (5/3) pi r^3

V_cyl = pi r^2 x r = pi r^3. V_hemi = (2/3) pi r^3. Total = pi r^3 + (2/3) pi r^3 = (5/3) pi r^3

H Challenge — Answer: 452.57 cm^3

Cylinder: r=6, h=12. V_cyl = pi x 36 x 12 = 1357.34. V_sphere = (4/3) x pi x 216 = 904.78. Space = 1357.34 - 904.78 = 452.57 cm^3

H Challenge — Answer: 50.27 cm^3

V_cyl = pi x 4 x 12 = 150.82. V_3_spheres = 3 x (4/3) x pi x 8 = 3 x 33.51 = 100.53. Space = 150.82 - 100.53 = 50.27 cm^3